

Measurement and quantum operations

What you can read out of a qubit, and what you are allowed to do to it.

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SESSION 12

Measurement

Postulate 3: how amplitudes turn into outcomes.

You never see the amplitudes. You see an outcome.

The amplitudes decide how likely each outcome is.

The Born rule, and what happens after

$$p(k) = |\langle b_k | \psi \rangle|^2$$

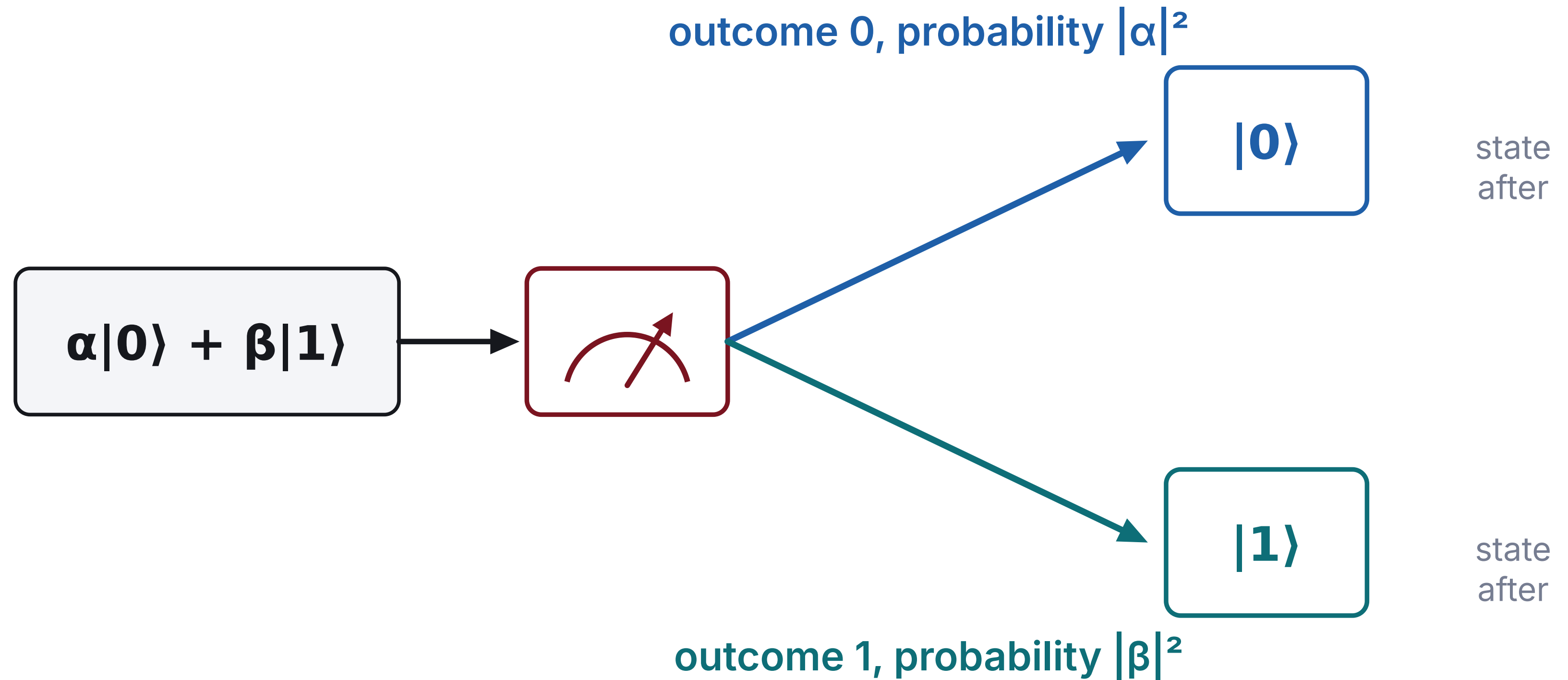
Measure $|\psi\rangle$ in an orthonormal basis $\{|b_k\rangle\}$. Outcome k comes with probability $p(k)$. Afterwards the state **is** $|b_k\rangle$: this is **collapse**.

Measuring in the basis $|0\rangle, |1\rangle$

$$p(0) = |\langle 0|\psi\rangle|^2 = |\alpha|^2, \quad p(1) = |\langle 1|\psi\rangle|^2 = |\beta|^2$$

For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. The two add to 1 because the state is normalised. That is why Postulate 1 insists on unit vectors.

The outcome is random. The state afterwards is not.



WORKED EXAMPLE 1

Complex amplitudes

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \frac{1+i}{\sqrt{3}}|1\rangle$$

$$p(0) = \left| \frac{1}{\sqrt{3}} \right|^2 = \frac{1}{3}, \quad p(1) = \frac{|1+i|^2}{3} = \frac{1^2 + 1^2}{3} = \frac{2}{3}$$

Check: $\frac{1}{3} + \frac{2}{3} = 1$. If the outcome is 1, the state afterwards is $|1\rangle$.

Square the modulus, never the amplitude

- **Squaring the amplitude**

$$\left(\frac{1+i}{\sqrt{3}}\right)^2 = \frac{2i}{3}$$

A “probability” of $2i/3$ is a sign you skipped the modulus.

- **Modulus, then square**

$$\left|\frac{1+i}{\sqrt{3}}\right|^2 = \frac{2}{3}$$

Always real, always between 0 and 1.

Measure twice, get the same answer twice

$$\alpha|0\rangle + \beta|1\rangle \xrightarrow{\text{outcome } 0} |0\rangle \xrightarrow{\text{measure again}} p(0) = |\langle 0|0\rangle|^2 = 1$$

The first measurement was random. Every repeat after it is certain. The original α and β are gone.

Measuring in the basis $|+\rangle, |-\rangle$

$$p(+)=|\langle +|\psi\rangle|^2,\qquad p(-)=|\langle -|\psi\rangle|^2$$

$$\langle \pm|\psi\rangle=\frac{1}{\sqrt{2}}\begin{pmatrix}1&\pm1\end{pmatrix}\begin{pmatrix}\alpha\\ \beta\end{pmatrix}=\frac{\alpha\pm\beta}{\sqrt{2}}$$

Same rule, different basis. The outcomes are now called "+" and "-".

WORKED EXAMPLE 2

Measure $|0\rangle$ in the $\pm$ basis

$$p(+)=\left|\frac{1+0}{\sqrt{2}}\right|^2=\frac{1}{2},\quad p(-)=\left|\frac{1-0}{\sqrt{2}}\right|^2=\frac{1}{2}$$

A state that is certain in one basis is a coin toss in another. After outcome "+", the state is $|+\rangle$.

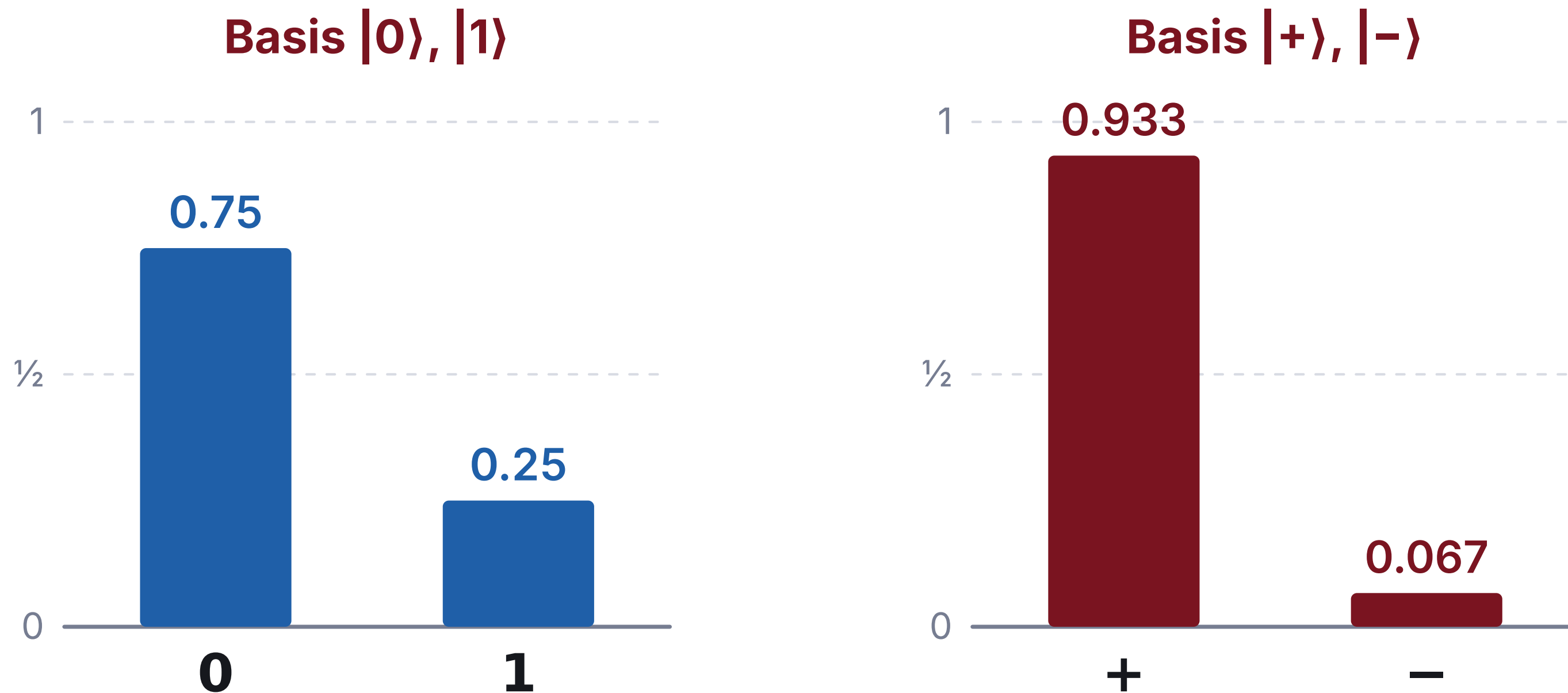
One state, two bases

$$|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle : \quad p(0) = \frac{3}{4}, \quad p(1) = \frac{1}{4}$$

$$\langle +|\psi\rangle = \frac{\sqrt{3} + 1}{2\sqrt{2}} \implies p(+)=\frac{(\sqrt{3} + 1)^2}{8} = \frac{4 + 2\sqrt{3}}{8} = \frac{2 + \sqrt{3}}{4} \approx 0.933$$

$$p(-)=\frac{(\sqrt{3} - 1)^2}{8} = \frac{2 - \sqrt{3}}{4} \approx 0.067$$

The basis you choose changes the odds



Same state, two measurements, two different sets of odds.

One measurement tells you almost nothing

0 or 1

1 copy: a single bit

75 $\pm$ 4

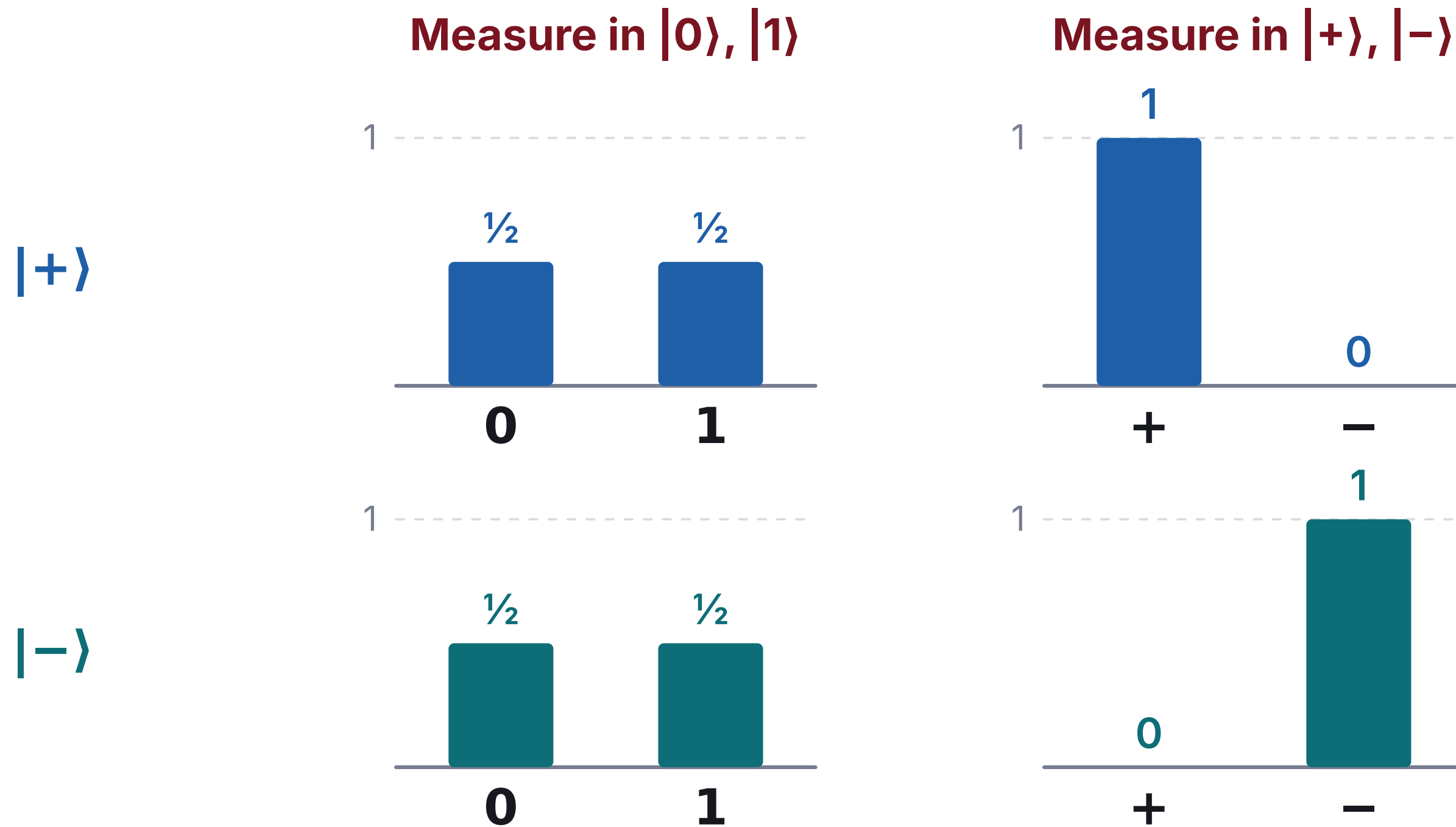
zeros in 100 copies

7500 $\pm$ 43

zeros in 10 000 copies

For $p(0) = \frac{3}{4}$. To estimate a probability you prepare the same state many times and count.

$|+\rangle$ and $|-\rangle$: identical in one basis, opposite in the other



TRY THIS NOW · 5 MINUTES

Measure $|+i\rangle$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i|1\rangle)$$

1. Find $p(0)$ and $p(1)$.
2. Find $p(+)$ and $p(-)$.
3. The outcome was “-”. What is the state now?

ANSWERS

A coin toss in both bases

1. $p(0) = p(1) = \frac{1}{2}$

2. $\langle +|\psi\rangle = \frac{1+i}{2}$, so $p(+)=\frac{2}{4}=\frac{1}{2}$. Likewise $p(-)=\left|\frac{1-i}{2}\right|^2=\frac{1}{2}$

3. $|-\rangle$

Only a measurement in the basis $|+i\rangle, |-i\rangle$ gives a certain answer for this state.

The Born rule is one line

```
import numpy as np
psi = np.array([1, 1j]) / np.sqrt(2)
plus = np.array([1, 1]) / np.sqrt(2)
minus = np.array([1, -1]) / np.sqrt(2)
print(abs(psi)**2)           # [0.5 0.5] basis 0, 1
print(abs(np.vdot(plus, psi))**2) # 0.5 outcome +
print(abs(np.vdot(minus, psi))**2) # 0.5 outcome -
```

What to do before next session

- Redo worked example 3 without looking, then check it in NumPy.
- Explain to a friend why $|+\rangle$ and $|-\rangle$ look the same in the basis $|0\rangle, |1\rangle$.
- Revise unitary matrices from session 8.

SESSION 13

Quantum operations

Postulate 2: what you are allowed to do to a state.

$$|\psi\rangle \rightarrow U|\psi\rangle$$

Between measurements, a closed system changes by a unitary matrix. A unitary acting on qubits is called a **gate**.

Two demands, one answer

stay normalised: $\langle \psi | U^\dagger U | \psi \rangle = 1$ for every $|\psi\rangle \implies U^\dagger U = I$

be undoable: $U^\dagger (U|\psi\rangle) = |\psi\rangle$

Probabilities must still add to 1 after the operation. Unitaries are exactly the matrices that guarantee it (session 8).

THE FIVE GATES OF THIS COURSE

X, Y, Z, H, S

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$$

Names: X is the bit flip, Z the phase flip, H the Hadamard gate, S the phase gate.

LEARN THIS TABLE

What each gate does to the standard states

GATE	ON $ 0\rangle$	ON $ 1\rangle$	ON $ +\rangle$	ON $ -\rangle$
X	$ 1\rangle$	$ 0\rangle$	$ +\rangle$	$- -\rangle$
Y	$i 1\rangle$	$-i 0\rangle$	$-i -\rangle$	$i +\rangle$
Z	$ 0\rangle$	$- 1\rangle$	$ -\rangle$	$ +\rangle$
H	$ +\rangle$	$ -\rangle$	$ 0\rangle$	$ 1\rangle$
S	$ 0\rangle$	$i 1\rangle$	$ +i\rangle$	$ -i\rangle$

WORKED EXAMPLE 4

X and Z on a general state

$$X \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \beta \\ \alpha \end{pmatrix} \qquad Z \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha \\ -\beta \end{pmatrix}$$

So X swaps the two amplitudes, while Z leaves both probabilities alone and puts a relative phase of $-1 = e^{i\pi}$ on $|1\rangle$.

The Hadamard gate, step by step

$$H|0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle$$

$$H^\dagger H = H^2 = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = I$$

The Hadamard gate is Hermitian and unitary, so it is its own inverse.

Apply A , then B : write BA

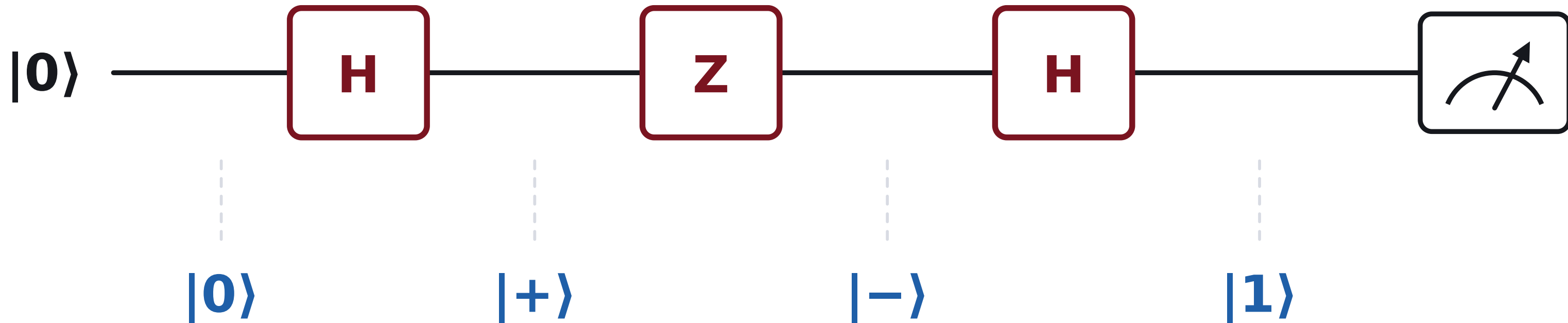
$$|\psi\rangle \rightarrow A|\psi\rangle \rightarrow B(A|\psi\rangle) = BA|\psi\rangle$$

The gate applied first sits next to the ket, on the right. Circuits read left to right. Matrix products read right to left.

A THREE-GATE CIRCUIT

H , then Z , then H : the matrix is HZH

Read the circuit left to right



WORKED EXAMPLE 6

$HZH|0\rangle$, one gate at a time

$$H|0\rangle = |+\rangle \quad \rightarrow \quad Z|+\rangle = |-\rangle \quad \rightarrow \quad H|-\rangle = |1\rangle$$

$$HZ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \quad HZH = \frac{1}{2} \begin{pmatrix} 1-1 & 1+1 \\ 1+1 & 1-1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = X$$

Both routes agree: sandwiching Z between two H gates gives X .

"Apply H, then S" to $|0\rangle$

- Writing the gates in time order

$$HS|0\rangle = H|0\rangle = |+\rangle$$

This applied S first.

- Rightmost acts first

$$SH|0\rangle = S|+\rangle = |+i\rangle$$

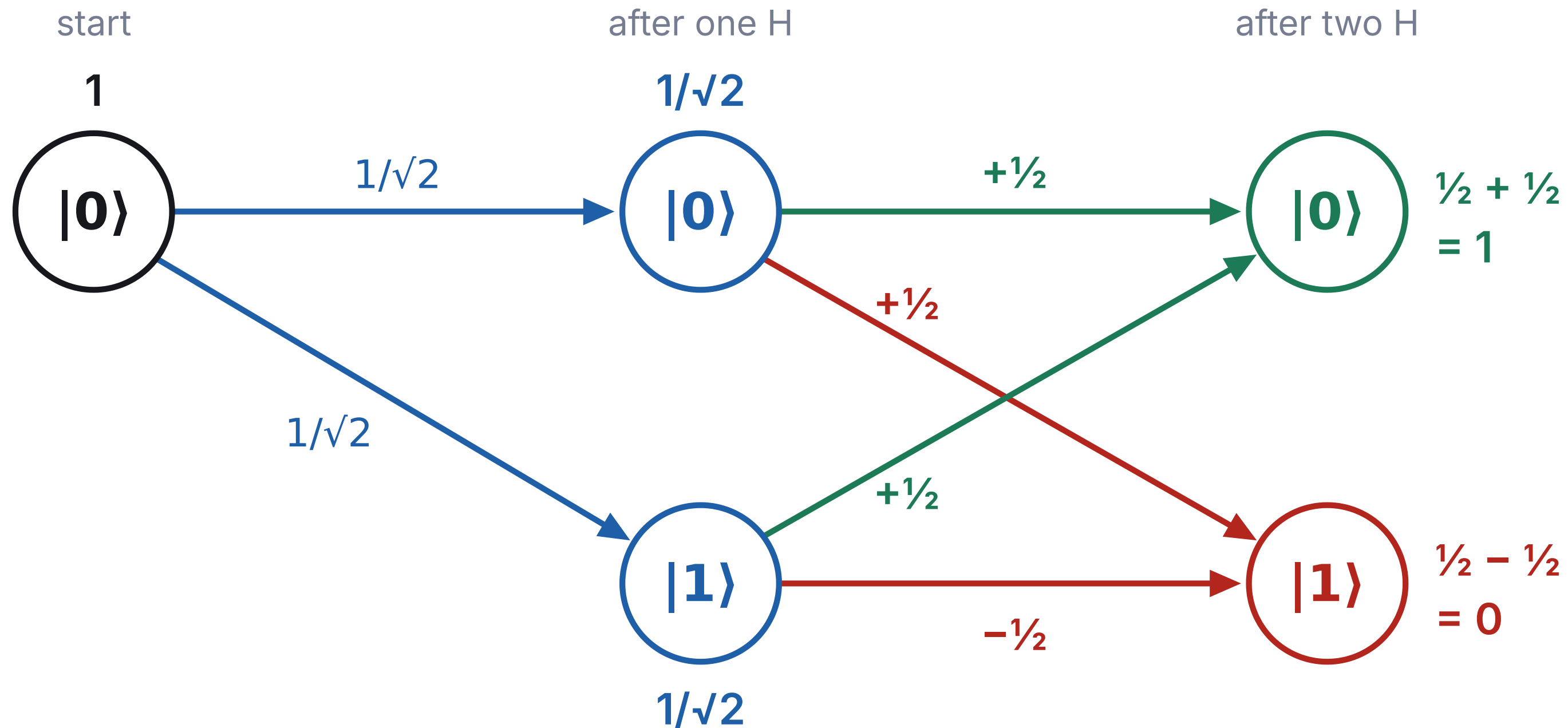
Different state. Gates do not commute in general.

Apply H twice and you are back at $|0\rangle$.

Amplitudes can cancel. Probabilities never do.

INTERFERENCE

Two paths to $|1\rangle$ cancel. Two paths to $|0\rangle$ add.



WORKED EXAMPLE 7

$HH|0\rangle$ in algebra

$$\begin{aligned} H(H|0\rangle) &= H|+\rangle = \frac{1}{\sqrt{2}} (H|0\rangle + H|1\rangle) \\ &= \frac{1}{2} (|0\rangle + |1\rangle + |0\rangle - |1\rangle) = \frac{1}{2} (2|0\rangle + 0|1\rangle) = |0\rangle \end{aligned}$$

The $+\frac{1}{2}$ and $-\frac{1}{2}$ on $|1\rangle$ cancel. That cancellation is **interference**.

Measuring between the two H gates

- **No measurement in between**

$$|0\rangle \rightarrow |+\rangle \rightarrow |0\rangle$$

Final outcome: 0, every time.

- **Measure after the first H**

Collapse to $|0\rangle$ or $|1\rangle$, each with probability $\frac{1}{2}$. The second H gives $|+\rangle$ or $|-\rangle$.

Final outcome: 0 or 1, fifty-fifty.

TRY THIS NOW · 5 MINUTES

Three short sequences

1. Compute $ZX|0\rangle$ and $XZ|0\rangle$. Same vector? Same physical state?
2. Compute $HXH|0\rangle$. Which single gate is HXH ?
3. Write S^2 as a matrix. Which gate is it?

Check yours

1. $ZX|0\rangle = Z|1\rangle = -|1\rangle$ and $XZ|0\rangle = X|0\rangle = |1\rangle$. Different vectors, same physical state: they differ by a global phase -1 .
2. $HXH|0\rangle = HX|+\rangle = H|+\rangle = |0\rangle$, and $HXH = Z$.
3. $S^2 = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = Z$

Gates are arrays, sequences are products

```
import numpy as np
X = np.array([[0, 1], [1, 0]])
Z = np.array([[1, 0], [0, -1]])
H = np.array([[1, 1], [1, -1]]) / np.sqrt(2)
S = np.array([[1, 0], [0, 1j]])
print(np.round(H @ Z @ H, 10))    # X
print(S @ S)                      # Z
ket0 = np.array([1, 0])
print(S @ H @ ket0)               # H first, then S: |+i>
```

What to do before next session

- Learn the gate table by heart. The end-semester exam assumes it.
- Assignment Part C: code the five gates and apply them to the six standard states.
- Revise the tensor product from session 9.